B3 Quantum, Atomic and Molecular Physics — Model Solutions, 2016

Question 1 — Alkali-like spectra of Ca^+

(a) Notation and selection rules

Problem. Explain the term symbol $3d\,^2D_{5/2} - 5p\,^2P_{3/2}$ and state the electric-dipole selection rules for alkali-like atoms.

For each level $n\ell\,^{2S+1}L_J$ of the single valence electron:

- n — principal quantum number of the active electron (3 and 5).
- ℓ — orbital quantum number of the active electron, given by the lowercase letter (d $\Rightarrow \ell = 2$, p $\Rightarrow \ell = 1$).
- $2S + 1 = 2$ — spin multiplicity, so $S = \frac{1}{2}$ (single valence electron).
- L — total orbital quantum number; for one electron $L = \ell$ (D$\Rightarrow L = 2$, P$\Rightarrow L = 1$).
- J — total electronic angular-momentum quantum number, $\mathbf{J} = \mathbf{L} + \mathbf{S}$, with values $J = 5/2$ and $J = 3/2$.

Electric-dipole selection rules (one-electron jump):

$$\Delta\ell = \pm1, \qquad \Delta L = \pm1, \qquad \Delta S = 0, \qquad \Delta J = 0, \pm1 \text{ (not } 0 \to 0), \qquad \Delta M_J = 0, \pm1.$$

For the quoted line $\Delta\ell = 2 - 1 = 1$ and $\Delta J = 5/2 - 3/2 = 1$: allowed.

(b) Energy-level diagram for Ca^+

Problem. Build an energy-level diagram from the doublet series: (i) resonance doublet at 393.4 nm, 396.9 nm; series limit 104.4 nm. (ii) doublet at 373.7 nm, 370.6 nm; series limit ~142 nm.

Ground configuration of Ca^+: [Ar] $4s\,^2S_{1/2}$.

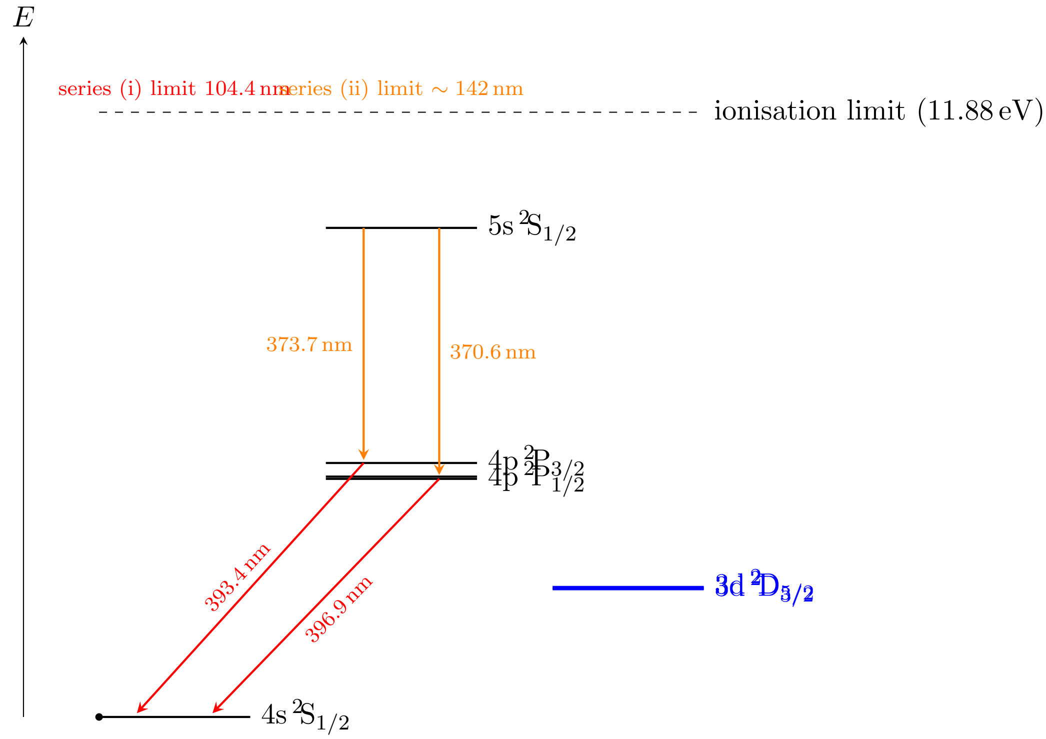
Series (i): the resonance lines start from the ground state $4s\,^2S_{1/2}$, so the first doublet is $4p\,^2P_{1/2,3/2} \to 4s\,^2S_{1/2}$. Energy of the series limit gives the ionisation energy from 4s:

$$E_{\text{ion}}(4s) = \frac{hc}{\lambda_\infty} = \frac{1239.8\,\text{eV nm}}{104.4\,\text{nm}} = 11.88\,\text{eV}.$$

Series (ii): its limit lies at ~ 142 nm, lower binding energy than (i):

$$E_{\text{ion}} = \frac{1239.8}{142.0} = 8.73\,\text{eV}.$$

This is consistent with the limit being reached from a more weakly bound lower term. The first doublet at 373.7 nm and 370.6 nm therefore originates from $5s\,^2S_{1/2} \to 4p\,^2P_{3/2,1/2}$ (allowed: $\Delta\ell = \pm1$). Equivalently, the lower term of series (ii) is $4p\,^2P$, which sits $E_{\text{ion}}(4s) - E_{\text{ion}}(4p) = 11.88 - 8.73 = 3.15\,\text{eV}$ above the ground state, matching $hc/393.4\,\text{nm} = 3.15\,\text{eV}$. Self-consistent.



(c) Triplet-IR series

Problem. Series limit at 121.8 nm; lines at 849.8 nm, 854.2 nm and 866.2 nm. Identify origin.

Ionisation energy from the lower term of this series:

$$E_{\text{ion}} = \frac{1239.8}{121.8} = 10.18\,\text{eV}.$$

Energy of lower term above ground:

$$11.88 - 10.18 = 1.70\,\text{eV}.$$

This matches the metastable 3d ^{2}D term (fine-structure doublet $J = 3/2,\ 5/2$). The IR lines therefore come from 4p ^{2}P$_{1/2,3/2}$ $\rightarrow$ 3d ^{2}D$_{3/2,5/2}$. Allowed combinations by $\Delta J = 0, \pm 1$ (excluding P$_{1/2} \rightarrow$ D$_{5/2}$ with $\Delta J = 2$).

Order by energy: P$_{3/2}$ above P$_{1/2}$; D$_{5/2}$ above D$_{3/2}$. Largest ΔE (shortest λ) is P$_{3/2} \rightarrow$ D$_{3/2}$; smallest ΔE is P$_{1/2} \rightarrow$ D$_{3/2}$. Therefore:

$$\boxed{\begin{aligned} &849.8\,\text{nm}: \ 4\text{p}\,{}^2\text{P}_{3/2} \rightarrow 3\text{d}\,{}^2\text{D}_{3/2}, \\ &854.2\,\text{nm}: \ 4\text{p}\,{}^2\text{P}_{3/2} \rightarrow 3\text{d}\,{}^2\text{D}_{5/2}, \\ &866.2\,\text{nm}: \ 4\text{p}\,{}^2\text{P}_{1/2} \rightarrow 3\text{d}\,{}^2\text{D}_{3/2}. \end{aligned}}$$

The transition P$_{1/2} \rightarrow$ D$_{5/2}$ ($\Delta J = 2$) is forbidden, hence only three lines.

(The 3d ^{2}D levels are added to the diagram above.)

(d) Zeeman pattern of the 393.4 nm line

Problem. Number of components observed perpendicular to $\mathbf{B}$ for 4p ^{2}P$_{3/2}$ $\rightarrow$ 4s ^{2}S$_{1/2}$.

In the anomalous Zeeman regime each level splits into $2J+1$ sublevels with shift $g_J\mu_B B M_J$:

$$g_J = 1 + \frac{J(J+1)+S(S+1)-L(L+1)}{2J(J+1)}.$$

Upper level $J=3/2$, $L=1$, $S=1/2$:

$$g_J = 1 + \frac{15/4+3/4-2}{2\cdot 15/4} = 1 + \frac{5/2}{15/2} = \tfrac{4}{3}.$$

Lower level $J=1/2$, $L=0$, $S=1/2$:

$$g_J = 2.$$

The g-factors differ, so the splitting is anomalous: each transition with $\Delta M_J = 0, \pm 1$ has a distinct frequency. Allowed transitions $M_J^{(u)} \to M_J^{(\ell)}$:

- $\Delta M_J = +1$: $-3/2 \to -1/2$, $-1/2 \to +1/2$ (σ^+, 2 lines).
- $\Delta M_J = 0$: $-1/2 \to -1/2$, $+1/2 \to +1/2$ (π, 2 lines).
- $\Delta M_J = -1$: $+3/2 \to +1/2$, $+1/2 \to -1/2$ (σ^-, 2 lines).

Total: 6 components, all distinct in frequency. Observed perpendicular to $\mathbf{B}$, π and $\sigma^\pm$ are all visible (as linearly polarised lines).

$$\boxed{\text{6 components}}$$

Question 2 — Interval rule, hyperfine structure of Mn

(a) Lande interval rule

Problem. Show $E_J - E_{J-1} = AJ$ for $H = A\,\mathbf{J}_1\cdot\mathbf{J}_2$.

Square the coupling $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$:

$$\mathbf{J}^2 = \mathbf{J}_1^2 + \mathbf{J}_2^2 + 2\,\mathbf{J}_1\cdot\mathbf{J}_2.$$

Solve for the dot product:

$$\mathbf{J}_1\cdot\mathbf{J}_2 = \tfrac{1}{2}\left(\mathbf{J}^2 - \mathbf{J}_1^2 - \mathbf{J}_2^2\right).$$

Take the eigenvalue in a $|J_1, J_2; J, M\rangle$ state:

$$E_J = \tfrac{1}{2}A\hbar^2[J(J+1) - J_1(J_1+1) - J_2(J_2+1)].$$

Difference between adjacent J levels:

$$E_J - E_{J-1} = \tfrac{1}{2}A\hbar^2[J(J+1) - (J-1)J] = \tfrac{1}{2}A\hbar^2\cdot 2J.$$

$$\boxed{E_J - E_{J-1} = A\hbar^2 J \equiv AJ}$$

(absorbing $\hbar^2$ into A).

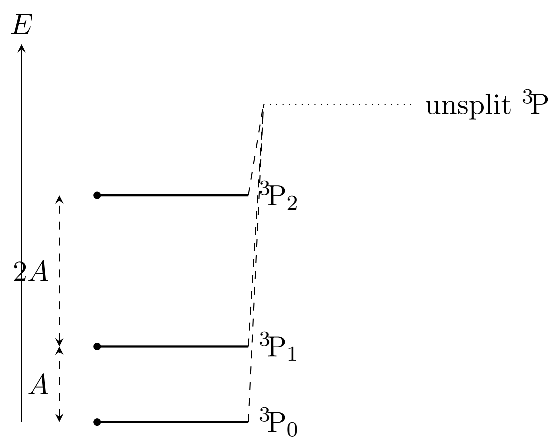
(b) Conditions for the rule and an example

Required conditions:

- Pure LS (Russell–Saunders) coupling: residual electrostatic interaction $\gg$ spin–orbit interaction, so L and S are good quantum numbers.
- Spin–orbit interaction is dominated by a single one-body term $\xi \mathbf{L}\cdot\mathbf{S}$ (or pair sum reducible to it), so $H_{SO} = A\,\mathbf{L}\cdot\mathbf{S}$.
- Higher-order effects (configuration interaction, spin–spin, magnetic interaction with neighbouring terms) are negligible.

Example: in the alkaline-earth two-electron atom such as Mg (3s3p ^{3}P), the term ^{3}P ($S = 1$, $L = 1$) splits into $J = 0, 1, 2$:

$$E_2 - E_1 = 2A, \qquad E_1 - E_0 = A, \qquad \frac{E_2 - E_1}{E_1 - E_0} = 2.$$



(c) Form of the hyperfine Hamiltonian

The electrons are in an eigenstate of $\mathbf{J}^2$ and J_z, so $\mathbf{B}_e$ at the nucleus is generated by the bound electrons and, by the Wigner–Eckart theorem (vector operator inside a manifold of fixed J), is proportional to $\mathbf{J}$:

$$\mathbf{B}_e = -\frac{B_e}{\hbar\sqrt{J(J+1)}}\mathbf{J} \propto \mathbf{J}.$$

Insert into H_{hfs}:

$$H_{\text{hfs}} = -g_I \mu_N \mathbf{I}\cdot\mathbf{B}_e = A\,\mathbf{I}\cdot\mathbf{J},$$

with A the hyperfine constant. Adding $\mathbf{F} = \mathbf{I} + \mathbf{J}$, the good quantum numbers are

$$\boxed{I,\ J,\ F,\ M_F}$$

(plus the configuration labels L, S that fix J).

(d) ^{55}Mn: deduce I, J, A

Problem. Splittings (MHz): 72, 145, 217.3, 289.7, 362.1.

By the interval rule the spacing between adjacent F levels is AF, so consecutive splittings form an arithmetic progression with common difference A:

$$\Delta\nu_F = AF.$$

Tabulate ratios:

$$72 : 145 : 217.3 : 289.7 : 362.1 \;\approx\; 1 : 2 : 3 : 4 : 5.$$

Therefore F runs over five consecutive integers (or half-integers) $F_{\min}, \ldots, F_{\max}$ with the smallest interval $AF_{\min} = 72$ MHz. Number of distinct F values is 6 (five intervals between six adjacent levels):

$$2\min(I, J) + 1 = 6 \quad\Longrightarrow\quad \min(I, J) = \tfrac{5}{2}.$$

The smallest $F = |I - J|$, and $AF_{\min} = 72$ MHz with the next interval $AF_{\min+1} = 145 \approx 2 \times 72$:

$$\frac{145}{72} = \frac{F_{\min} + 1}{F_{\min}} \Rightarrow F_{\min} = 1.$$

Cross-check: ratios $217.3/72 = 3.02$, $289.7/72 = 4.02$, $362.1/72 = 5.03$ — consistent. So $F_{\min} = 1$, $F_{\max} = 6$. With $\min(I, J) = 5/2$ and $F_{\min} = |I - J| = 1$:

$$|I - J| = 1, \qquad I + J = 6,$$

giving $\{I, J\} = \{5/2,\, 7/2\}$.

The ground configuration of Mn is [Ar]$3d^54s^2$; Hund's rules give a half-filled $3d^5$ shell, $L = 0$, $S = 5/2$, hence $J = 5/2$ ($^6S_{5/2}$). Therefore:

$$\boxed{J = \tfrac{5}{2}, \quad I(^{55}\mathrm{Mn}) = \tfrac{5}{2}}$$

With $AF_{\min} = A \cdot 1 = 72$ MHz:

$$\boxed{A(^{55}\mathrm{Mn}) = 72.4\,\mathrm{MHz}}$$

(taking the average $A = (72 + 145/2 + 217.3/3 + 289.7/4 + 362.1/5)/5 \approx 72.4$ MHz).

(e) ^{56}Mn: ratio of nuclear g-factors

Splittings: 85, 141, 197.4, 253.8, 310.2 MHz. Differences:

$$141 - 85 = 56, \quad 197.4 - 141 = 56.4, \quad 253.8 - 197.4 = 56.4, \quad 310.2 - 253.8 = 56.4.$$

Common difference $A^{(56)} = 56.4$ MHz. The smallest splitting is at $F_{\min}$:

$$A^{(56)} F_{\min} = 85 \;\Rightarrow\; F_{\min} = \frac{85}{56.4} \approx 1.51 \approx \tfrac{3}{2}.$$

There are again 5 intervals $\Rightarrow$ 6 adjacent levels $\Rightarrow$ $2\min(I, J) + 1 = 6 \Rightarrow \min(I, J) = 5/2$. With $J = 5/2$ unchanged and $F_{\min} = |I - J| = 3/2$:

$$I(^{56}\mathrm{Mn}) = J + F_{\min} = \tfrac{5}{2} + \tfrac{3}{2} = 4 \quad \text{(or } I = 1,\text{ but the spread of 5 intervals fixes } 2\min(I, J) = 5\text{).}$$

Take $I = 4$ (the odd-odd nucleus has the larger spin; only this gives 6 adjacent F levels). F ranges $|I - J|, \ldots, I + J = \frac{3}{2}, \ldots, \frac{13}{2}$: 6 levels, 5 intervals.

Ratio of magnetic moments. The hyperfine constant satisfies

$$A = \frac{g_I \mu_N B_e}{\hbar^2 \sqrt{J(J+1)}\sqrt{I(I+1)}} \cdot \hbar^2 \;\propto\; g_I,$$

since J and the electronic configuration (hence B_e) are common to the two isotopes. Therefore

$$\frac{g_I(^{56}\mathrm{Mn})}{g_I(^{55}\mathrm{Mn})} = \frac{A^{(56)}}{A^{(55)}} = \frac{56.4}{72.4}.$$

$$\boxed{\frac{g_I(^{56}\mathrm{Mn})}{g_I(^{55}\mathrm{Mn})} \approx 0.78}$$

Question 3 — Three- and four-level lasers, gain saturation

(a) Three- vs. four-level lasers

Three-level laser: pump $|0\rangle \to |2\rangle$, fast non-radiative decay $|2\rangle \to |1\rangle$, lasing transition $|1\rangle \to |0\rangle$ (lower laser level = ground state). To achieve population inversion $N_1 > N_0/2$, more than half the atoms must be removed from the ground state, requiring large pump power. Example: ruby (Cr^{3+}:Al_2O_3, $\lambda = 694.3\,\mathrm{nm}$).

Four-level laser: pump $|0\rangle \to |3\rangle$, fast decay $|3\rangle \to |2\rangle$, lasing $|2\rangle \to |1\rangle$, fast decay $|1\rangle \to |0\rangle$. Lower laser level is essentially empty in the steady state, so any population in $|2\rangle$ provides inversion. Example: Nd:YAG (1064 nm), He–Ne (632.8 nm).

Threshold: threshold inversion $N^*_{\mathrm{th}} = \gamma/\sigma$ (γ = round-trip loss). For three-level $N^*_{\mathrm{th}} \approx N_0/2$, pump rate $\propto N_{\mathrm{tot}}/2$. For four-level N^*_{th} may be many orders of magnitude smaller than N_{tot}, so threshold pump power is correspondingly smaller.

(b) Rate equations without radiation

Problem. Steady-state inversion with pump R_2, decay A_{21} only, lower-level lifetime τ_1 fed only by spontaneous emission.

Rate equations:

$$\dot{N}_2 = R_2 - A_{21}N_2,$$
$$\dot{N}_1 = A_{21}N_2 - \frac{N_1}{\tau_1}.$$

Steady state $\dot{N}_2 = \dot{N}_1 = 0$:

$$N_2 = \frac{R_2}{A_{21}}, \qquad N_1 = A_{21}N_2\tau_1 = R_2\tau_1.$$

Inversion density:

$$N^* = N_2 - \frac{g_2}{g_1}N_1 = \frac{R_2}{A_{21}} - \frac{g_2}{g_1}R_2\tau_1.$$

$$\boxed{N^* = R_2\left(\frac{1}{A_{21}} - \frac{g_2}{g_1}\tau_1\right)}$$

(c) Rate equations with radiation; saturation

Add stimulated emission/absorption. Define $\sigma(\omega)$ = stimulated cross-section; the stimulated rate per atom is $\sigma I/\hbar\omega$. Including degeneracy ratio g_2/g_1:

$$\dot{N}_2 = R_2 - A_{21}N_2 - \frac{\sigma I}{\hbar\omega}\left(N_2 - \frac{g_2}{g_1}N_1\right),$$

$$\dot{N}_1 = A_{21}N_2 - \frac{N_1}{\tau_1} + \frac{\sigma I}{\hbar\omega}\left(N_2 - \frac{g_2}{g_1}N_1\right).$$

Define stimulated rate $W \equiv \sigma I/\hbar\omega$. Steady state:

$$N_2 = \frac{R_2}{A_{21} + W(1 - g_2N_1/(g_1N_2))},$$

better: combine the two equations to obtain N^* directly. Form $\dot{N}_2 - (g_2/g_1)\dot{N}_1 = 0$:

$$0 = R_2 - A_{21}N_2 - WN^* - \frac{g_2}{g_1}\left(A_{21}N_2 - \frac{N_1}{\tau_1} + WN^*\right).$$

Group terms:

$$0 = R_2 - A_{21}N_2\left(1 + \frac{g_2}{g_1}\right) + \frac{g_2N_1}{g_1\tau_1} - WN^*\left(1 + \frac{g_2}{g_1}\right).$$

Use $\dot{N}_2 = 0$ to write $N_2 = (R_2 - WN^*)/A_{21}$ and $\dot{N}_1 = 0 \Rightarrow N_1 = \tau_1(A_{21}N_2 + WN^*)$:

$$N^* = N_2 - \frac{g_2}{g_1}N_1 = \frac{R_2 - WN^*}{A_{21}} - \frac{g_2}{g_1}\tau_1(A_{21}N_2 + WN^*).$$

Substitute $A_{21}N_2 = R_2 - WN^*$:

$$N^* = \frac{R_2 - WN^*}{A_{21}} - \frac{g_2}{g_1}\tau_1(R_2 - WN^* + WN^*).$$

Simplify the last bracket:

$$N^* = \frac{R_2 - WN^*}{A_{21}} - \frac{g_2}{g_1}\tau_1R_2.$$

Rearrange:

$$N^*\left(1 + \frac{W}{A_{21}}\right) = \frac{R_2}{A_{21}} - \frac{g_2}{g_1}\tau_1R_2 = N_0^*.$$

Define the unsaturated inversion $N_0^* \equiv R_2(1/A_{21} - (g_2/g_1)\tau_1)$:

$$N^* = \frac{N_0^*}{1 + W/A_{21}}.$$

Gain coefficient $\alpha = \sigma N^*$:

$$\alpha = \frac{\sigma N_0^*}{1 + \sigma I/(\hbar\omega A_{21})}.$$

$$\boxed{\alpha = \frac{\alpha_0}{1 + I/I_s}, \qquad \alpha_0 = \sigma(\omega_{21})N_0^*, \qquad I_s = \frac{\hbar\omega_{21}A_{21}}{\sigma(\omega_{21})}.}$$

α_0: small-signal (unsaturated) gain coefficient. I_s: saturation intensity — the intensity at which α falls to $\alpha_0/2$.

Limit $I \gg I_s$. Then $\alpha \approx \alpha_0I_s/I$, so

$$\frac{\mathrm{d}I}{\mathrm{d}z} = \alpha I = \alpha_0I_s.$$

Intensity grows *linearly* with z (saturated amplifier):

$$I(z) = I(0) + \alpha_0I_sz.$$

(d) Numerical α_0

Transition 1: $\sigma = 4 \times 10^{-15}\,\mathrm{cm}^2$, $R_2 = 5 \times 10^{19}\,\mathrm{s}^{-1}\mathrm{cm}^{-3}$, $A_{21} = 3 \times 10^6\,\mathrm{s}^{-1}$, $g_2 = 5$, $\tau_1 = 2 \times 10^{-8}\,\mathrm{s}$, $g_1 = 3$.

Compute N_0^* piecewise:

$$\frac{1}{A_{21}} = \frac{1}{3 \times 10^6} = 3.33 \times 10^{-7}\,\mathrm{s}.$$

$$\frac{g_2}{g_1}\tau_1 = \frac{5}{3} \times 2 \times 10^{-8} = 3.33 \times 10^{-8}\,\mathrm{s}.$$

Difference:

$$\frac{1}{A_{21}} - \frac{g_2}{g_1}\tau_1 = 3.33 \times 10^{-7} - 0.333 \times 10^{-7} = 3.00 \times 10^{-7}\,\mathrm{s}.$$

Multiply by R_2:

$$N_0^* = 5 \times 10^{19} \times 3.00 \times 10^{-7} = 1.50 \times 10^{13}\,\mathrm{cm}^{-3}.$$

Multiply by σ:

$$\alpha_0 = 4 \times 10^{-15} \times 1.50 \times 10^{13} = 6.0 \times 10^{-2}\,\mathrm{cm}^{-1}.$$

$$\boxed{\alpha_0^{(1)} \approx 0.060\,\mathrm{cm}^{-1}}$$

Transition 2: $A_{21} = 3 \times 10^6\,\mathrm{s}^{-1}$, $g_2 = 3$, $\tau_1 = 3 \times 10^{-7}\,\mathrm{s}$, $g_1 = 1$, same σ, R_2.

$$\frac{1}{A_{21}} = 3.33 \times 10^{-7}\,\mathrm{s}.$$

$$\frac{g_2}{g_1}\tau_1 = 3 \times 3 \times 10^{-7} = 9.0 \times 10^{-7}\,\mathrm{s}.$$

Difference:

$$3.33 \times 10^{-7} - 9.0 \times 10^{-7} = -5.67 \times 10^{-7}\,\mathrm{s}.$$

Multiply by R_2:

$$N_0^* = 5 \times 10^{19} \times (-5.67 \times 10^{-7}) = -2.83 \times 10^{13}\,\mathrm{cm}^{-3}.$$

$$\boxed{\alpha_0^{(2)} \approx -0.113\,\mathrm{cm}^{-1}}$$

$N_0^* < 0$: no inversion ever (the lower level lives too long), so this transition cannot lase.

Question 4 — Two-state atom in an oscillating field

(a) Coupled amplitude equation

Problem. Derive $\dot{c}_2 = -i\Omega\cos(\omega t)\mathrm{e}^{i\omega_0 t}c_1 - (iV_{22}/\hbar)c_2$ from the TDSE.

Insert $\Psi = c_1\psi_1\mathrm{e}^{-iE_1t/\hbar} + c_2\psi_2\mathrm{e}^{-iE_2t/\hbar}$ into $i\hbar\partial_t\Psi = (\hat{H}_0 + \hat{V})\Psi$:

$$i\hbar\Big(\dot{c}_1\psi_1\mathrm{e}^{-iE_1t/\hbar} + \dot{c}_2\psi_2\mathrm{e}^{-iE_2t/\hbar}\Big) = \hat{V}\Big(c_1\psi_1\mathrm{e}^{-iE_1t/\hbar} + c_2\psi_2\mathrm{e}^{-iE_2t/\hbar}\Big).$$

Project onto ψ_2 (multiply by ψ_2^* and integrate):

$$i\hbar\dot{c}_2\mathrm{e}^{-iE_2t/\hbar} = c_1V_{21}\mathrm{e}^{-iE_1t/\hbar} + c_2V_{22}\mathrm{e}^{-iE_2t/\hbar},$$

where $V_{ij} = \langle\psi_i|\,\hat{V}(t)\,|\psi_j\rangle = \langle\psi_i|\,ex\,|\psi_j\rangle\,\mathcal{E}\cos\omega t$. Multiply both sides by $\mathrm{e}^{iE_2t/\hbar}/(i\hbar)$:

$$\dot{c}_2 = \frac{V_{21}}{i\hbar}\mathrm{e}^{i(E_2-E_1)t/\hbar}c_1 + \frac{V_{22}}{i\hbar}c_2.$$

Define $\omega_0 \equiv (E_2 - E_1)/\hbar$ and the Rabi frequency

$$\Omega \equiv \frac{e\mathcal{E}}{\hbar} \langle\psi_2| x |\psi_1\rangle = \frac{\langle\psi_2| ex |\psi_1\rangle \mathcal{E}}{\hbar}.$$

Then $V_{21} = \hbar\Omega\cos(\omega t)$:

$$\boxed{\dot{c}_2 = -i\Omega\cos(\omega t)\mathrm{e}^{i\omega_0 t}c_1 - \frac{iV_{22}}{\hbar}c_2}$$

with

$$\omega_0 = \frac{E_2 - E_1}{\hbar}, \qquad V_{22} = \langle\psi_2| ex |\psi_2\rangle \mathcal{E}\cos\omega t, \qquad \Omega = \frac{e\mathcal{E}\langle\psi_2| x |\psi_1\rangle}{\hbar}.$$

Why $V_{22} = 0$. The atomic eigenstates ψ_2 have definite parity (since $\hat{H}_0$ is parity-even, eigenfunctions of a non-degenerate level have definite parity); $|\psi_2|^2$ is even and x is odd, so $\int |\psi_2|^2 x\,\mathrm{d}^3\mathbf{r} = 0$. Hence $\langle\psi_2| x |\psi_2\rangle = 0$ and $V_{22} = 0$.

(b) First-order $c_2(t)$

With $V_{22} = 0$, $c_1 \simeq 1$, and $\cos(\omega t) = \frac{1}{2}(\mathrm{e}^{i\omega t} + \mathrm{e}^{-i\omega t})$:

$$\dot{c}_2 = -\frac{i\Omega}{2}\left[\mathrm{e}^{i(\omega_0+\omega)t} + \mathrm{e}^{i(\omega_0-\omega)t}\right].$$

Integrate from 0 to t with $c_2(0) = 0$:

$$c_2(t) = -\frac{i\Omega}{2}\left[\frac{\mathrm{e}^{i(\omega_0+\omega)t} - 1}{i(\omega_0+\omega)} + \frac{\mathrm{e}^{i(\omega_0-\omega)t} - 1}{i(\omega_0-\omega)}\right].$$

$$\boxed{c_2(t) = -\frac{\Omega}{2}\left[\frac{\mathrm{e}^{i(\omega_0+\omega)t} - 1}{\omega_0+\omega} + \frac{\mathrm{e}^{i(\omega_0-\omega)t} - 1}{\omega_0-\omega}\right]}$$

(c) Rotating-wave approximation; $|c_2|^2$

Near resonance $\omega \sim \omega_0$: the term $1/(\omega_0-\omega)$ is small in denominator and dominates, while $1/(\omega_0+\omega) \sim 1/(2\omega_0)$ is small and oscillates rapidly — it averages to zero on the timescale of interest. Drop the counter-rotating term (RWA):

$$c_2(t) \approx -\frac{\Omega}{2}\frac{\mathrm{e}^{i(\omega_0-\omega)t} - 1}{\omega_0-\omega}.$$

Factor out $\mathrm{e}^{i(\omega_0-\omega)t/2}$:

$$c_2(t) = -\frac{\Omega}{2}\mathrm{e}^{i(\omega_0-\omega)t/2}\,\frac{2i\sin\frac{1}{2}(\omega_0-\omega)t}{\omega_0-\omega}.$$

Take the modulus squared:

$$\boxed{|c_2(t)|^2 = \Omega^2\frac{\sin^2\frac{1}{2}(\omega_0-\omega)t}{(\omega_0-\omega)^2} = \frac{\Omega^2 t^2}{4}\,\mathrm{sinc}^2\!\left(\frac{(\omega_0-\omega)t}{2}\right)}$$

(d) Broadband stimulated absorption

For broadband radiation spectral energy density $\rho(\omega)$ replaces $\frac{1}{2}\epsilon_0\mathcal{E}^2$. The Rabi frequency squared at frequency ω is

$$\Omega^2(\omega) = \frac{e^2|\langle\psi_2|\,x\,|\psi_1\rangle|^2\mathcal{E}^2(\omega)}{\hbar^2} = \frac{2e^2|x_{12}|^2}{\epsilon_0\hbar^2}\rho(\omega)\,\mathrm{d}\omega.$$

Different frequency components are mutually incoherent; sum (integrate) the probabilities:

$$|c_2(t)|^2 = \int_0^\infty \frac{2e^2|x_{12}|^2}{\epsilon_0\hbar^2}\rho(\omega)\,\frac{\sin^2\frac{1}{2}(\omega_0-\omega)t}{(\omega_0-\omega)^2}\,\mathrm{d}\omega.$$

For weak slowly-varying ρ around ω_0, pull $\rho(\omega_0)$ out of the integral and extend limits to $\pm\infty$:

$$|c_2(t)|^2 \approx \frac{2e^2|x_{12}|^2}{\epsilon_0\hbar^2}\rho(\omega_0)\int_{-\infty}^{\infty}\frac{\sin^2\frac{1}{2}(\omega_0-\omega)t}{(\omega_0-\omega)^2}\,\mathrm{d}\omega.$$

Apply the given integral $\int_{-\infty}^{\infty}[\sin\frac{1}{2}(\omega-\omega_0)t/\frac{1}{2}(\omega-\omega_0)]^2\mathrm{d}\omega = 2\pi t$, equivalently

$$\int_{-\infty}^{\infty}\frac{\sin^2\frac{1}{2}(\omega_0-\omega)t}{(\omega_0-\omega)^2}\,\mathrm{d}\omega = \frac{\pi t}{2}\cdot 2 = \frac{\pi t}{2}\cdot 2 \;\Rightarrow\; \frac{\pi t}{2}\cdot 2.$$

Carefully: with $u = (\omega-\omega_0)t/2$, $\sin^2 u/u^2\cdot(t/2)\mathrm{d}u/((2u/t)^2)\cdot$ — use the supplied result directly:

$$\int_{-\infty}^{\infty}\frac{\sin^2\frac{1}{2}(\omega-\omega_0)t}{[\frac{1}{2}(\omega-\omega_0)]^2}\,\mathrm{d}\omega = 2\pi t \;\Rightarrow\; \int_{-\infty}^{\infty}\frac{\sin^2\frac{1}{2}(\omega-\omega_0)t}{(\omega-\omega_0)^2}\,\mathrm{d}\omega = \frac{\pi t}{2}.$$

Hence

$$|c_2(t)|^2 = \frac{2e^2|x_{12}|^2}{\epsilon_0\hbar^2}\rho(\omega_0)\cdot\frac{\pi t}{2} = \frac{\pi e^2|x_{12}|^2}{\epsilon_0\hbar^2}\rho(\omega_0)\,t.$$

Transition rate $W_{12} = \mathrm{d}|c_2|^2/\mathrm{d}t$:

$$\boxed{W_{12} = \frac{\pi e^2|\langle\psi_2|\,x\,|\psi_1\rangle|^2}{\epsilon_0\hbar^2}\rho(\omega_0)}$$

This is constant in time: standard Fermi-golden-rule form for stimulated absorption. (A factor 1/3 enters if averaging over polarisation directions and using $|\langle\psi_2|\,\mathbf{r}\,|\psi_1\rangle|^2$ instead of the x-component alone.)